# Tide retrospective: monomial variance positivity (weighted-jet programme, stage 1)

Tide loop (Claude Fable 5, with GPT-5.6 Sol deliberation)

2026-08-10

## 1 Setting

This tide opens the weighted-jet programme: the germbij note's §7.4 inductive recovery, in which each coefficient $c_r$ of a degenerate one-dimensional potential $L = a\,x^{2k} + \sum_r c_r x^{2k+r}$ is read off the graded correction at $t^{-r/(2k)}$ by a covariance pairing. The programme was scoped against three alternatives (the quintic cumulant rung, the general-order nondegenerate induction, and the multivariate first rung); the consult ranked the weighted jet first — smallest infrastructure delta per stage, closed forms at every rung, and its normalized-series machinery de-risks the general-order induction later. The named scope trap: the bare finite jet need not be globally integrable (the top term may be odd or negative), so the analytic envelope must be fixed before the combinatorics.

The seabed survey collapsed the consult's stage 1: the generalized-Gaussian moment API (partition function, even and odd moments, Gibbs expectations, integrability) already exists in `MonomialPotential.lean` from the recovery thread. What was missing is exactly the injectivity input every recovery rung will consume: strict positivity of the variance of a monomial observable against the reference weight.

## 2 The thing formalised

Three theorems in `Laplace/OneD/MonomialVariance.lean`, sorry-free with zero warnings, for the reference weight $e^{-t\,x^{2k}/(2k)!}$ with $k \geq 1$, $t > 0$: the Gamma closed forms of $\mathrm{Var}(x^n)$ for odd $n$ (the mean vanishes by symmetry, so the variance is the even moment)[1] and even $n$ (a difference of Gamma ratios under the common power of $(2k)!/t)^2$, and the programme's load-bearing fact[3]:

$$\mathrm{Var}_{\mu_{t,k}}(x^n) \;>\; 0 \qquad (n \geq 1).$$

## 3 Proof strategy

The positivity proof formalises the note's own injectivity argument — $\mathrm{Var}[Q] = 0$ would force $Q$ constant against a measure of full support. Concretely: with $M = \langle x^n \rangle$ and $Z$ the partition function,

$$\mathrm{Var}(x^n) = \frac{1}{Z} \int (x^n - M)^2 \, e^{-tL_k} \, dx,$$

[1] `monomial_variance_odd`.
[2] `monomial_variance_even`.
[3] `monomial_variance_pos`.

and the centered integrand is continuous, nonnegative, and not identically zero, so its support is open and nonempty (witness $x = 1$ when $M = 0$, else $x = 0$), hence of positive Lebesgue measure, and Mathlib's `integral_pos_iff_support_of_nonneg` gives positivity; dividing by $Z > 0$ finishes. The variance identity itself is the usual expansion $\int (x^n - M)^2 w = \int x^{2n} w - 2M \int x^n w + M^2 Z$ with the middle term collapsed by $\int x^n w = MZ$.

The numerical check (executed first) validated both the tide and the programme: the Gamma-form variances match quadrature to six decimals ($k \in \{1, 2\}$, $n \leq 4$), and — the consult's prescribed programme-level check — the $q^r$ coefficient of $\mathbb{E}_q[u^{2k+r}]$ responds to $c_r$ with slope $-\mathrm{Var}(u^{2k+r})$ independent of the lower coefficients (measured/predicted ratios 1.001–1.006 at $q = 0.05$, both parities of $2k + r$ included).

# 4 Roadblocks and resolutions

Three small ones, two of them new catalogue entries. First, `ring` *does* normalise symbolic exponent arithmetic: $x^{2n}$ versus $(x^n)^2$ with $n$ a free natural closes by plain `ring`, and the pre-planned `two_mul`/`pow_add` rewrites were actively harmful (`two_mul` matched the constant $2M$ first). Second, `simp only [defName]` does not unfold a definition — `unfold` does, after which `fun_prop` dispatches continuity of the monomial potential. Third, `positivity` cannot conclude $\neq 0$ through $(-M)^2$ without $M \neq 0$ in a form it consumes; explicit `mul_ne_zero` chains are cleaner for support witnesses.

# 5 The deliberation

The programme-level consult is archived verbatim in the tide log.[4] Both parties voted for the weighted-jet programme. Its stage list assigns this tide the reference moment API; the seabed survey showed most of that stage already merged, so the tide contracted to the variance layer — the kind of adjustment the survey step exists to catch. The consult's warning that partition data alone would lose odd perturbations at linear order (symmetry) is precisely why the variance against the *matching moment observable* $u^{2k+r}$ is the right injectivity quantity, and why this tide is stage 1 rather than an afterthought.

# 6 Mathlib vs new

Mathlib lookup: `integral_pos_iff_support_of_nonneg`, open-set measure positivity, preimage openness of the support of a continuous function. Seabed reuse: the entire moment API. New: the variance identity assembly and the support witnesses.

# 7 What the next tide inherits

Stage 2 of the programme: the exact scaling identity $t^{s/(2k)} \langle x^s \rangle_t$ as a normalized perturbation of the reference measure for the finite jet, with the analytic envelope chosen explicitly (the consult's trap warning). The variance layer here is consumed at stage 6–7, where the triangular covariance lemma pins each $c_r$ with denominator $\mathrm{Var}_\mu(u^{2k+r}) > 0$.

[4] `tide-log/gpt56_weighted_jet_scoping_v1.md`.